A First Course in Abstract Algebra

Section 23: Factorization of Polynomials over a Field

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

Contents

23 Factorization of Polynomials over a Field 3

Exercise 23.1: Polynomial division 1 3

Exercise 23.2: Polynomial division 2 3

Exercise 23.3: Polynomial division 3 3

Exercise 23.4: Polynomial division 4 4

Exercise 23.5: All multiplicative generators modulo 5 4

Exercise 23.6: All multiplicative generators modulo 7 4

Exercise 23.7: All multiplicative generators modulo 17 4

Exercise 23.8: All multiplicative generators modulo 23 4

Exercise 23.9: Four linear factors modulo five 5

Exercise 23.10: Three linear factors modulo seven 5

Exercise 23.11: A nonmonic cubic modulo eleven 5

Exercise 23.12: A reducible cubic modulo five 5

Exercise 23.13: An irreducible cubic modulo five 5

Exercise 23.14: A quadratic over three fields 6

Exercise 23.15: A quadratic with negative discriminant 6

Exercise 23.16: A cubic over the rationals . . . 6
Exercise 23.17: A quartic with no quadratic factor . . . 6
Exercise 23.18: An Eisenstein quadratic . . . 6
Exercise 23.19: An Eisenstein cubic . . . 7
Exercise 23.20: An Eisenstein polynomial of degree ten . . . 7
Exercise 23.21: Another Eisenstein polynomial of degree ten . . . 7
Exercise 23.22: All rational zeros of a quartic . . . 7
Exercise 23.23: Correcting irreducibility . . . 7
Exercise 23.24: An equivalent irreducibility definition . . . 7
Exercise 23.25: True or false: irreducibility and root bounds . . . 7
Exercise 23.26: Primes for which a specified linear factor divides . . . 8
Exercise 23.27: Irreducible quadratics over two elements . . . 8
Exercise 23.28: Irreducible cubics over two elements . . . 8
Exercise 23.29: Irreducible quadratics over three elements . . . 8
Exercise 23.30: Irreducible cubics over three elements . . . 9
Exercise 23.31: Counting irreducible quadratics over a prime field . . . 9
Exercise 23.32: Synopsis of the bound on roots . . . 9
Exercise 23.33: Synopsis of cyclicity of finite field subgroups . . . 9
Exercise 23.34: A reducible prime-power polynomial . . . 9
Exercise 23.35: The reciprocal polynomial . . . 10
Exercise 23.36: The remainder theorem . . . 10
Exercise 23.37: Reduction of integer coefficients . . . 10

23 Factorization of Polynomials over a Field

Exercise 23.1: Polynomial division 1

Work in $\mathbb{Z}_7[x]$, reducing coefficients after each subtraction. The inverse of the divisor's leading coefficient is 1. Repeatedly subtract the multiple that cancels the current highest power. The complete long-division steps are

term added to quotient	new remainder
x^4	$x^5 + 3x^4 + 4x^2 + 4x + 2$
x^3	$x^4 + 3x^3 + 4x^2 + 4x + 2$
x^2	$x^3 + 4x + 2$
x	$5x^2 + 2$
5	$4x + 3$

Thus $\boxed{q(x) = x^4 + x^3 + x^2 + x + 5}$ and $\boxed{r(x) = 4x + 3}$. The final remainder has degree below the divisor's degree; reversing the subtractions verifies $f = gq + r$.

Exercise 23.2: Polynomial division 2

Work in $\mathbb{Z}_7[x]$, reducing coefficients after each subtraction. The inverse of the divisor's leading coefficient is 5. Repeatedly subtract the multiple that cancels the current highest power. The complete long-division steps are

term added to quotient	new remainder
$5x^4$	$x^4 + 4x^2 + 4x + 2$
$5x^2$	$4x^3 + 5x^2 + 4x + 2$
$6x$	$x + 2$

Thus $\boxed{q(x) = 5x^4 + 5x^2 + 6x}$ and $\boxed{r(x) = x + 2}$. The final remainder has degree below the divisor's degree; reversing the subtractions verifies $f = gq + r$.

Exercise 23.3: Polynomial division 3

Work in $\mathbb{Z}_{11}[x]$, reducing coefficients after each subtraction. The inverse of the divisor's leading coefficient is 6. Repeatedly subtract the multiple that cancels the current highest power. The complete long-division steps are

term added to quotient	new remainder
$6x^4$	$3x^4 + 3x + 6$
$7x^3$	$4x^3 + 3x + 6$
$2x^2$	$9x^2 + 3x + 6$
$10x$	$4x + 6$
2	4

Thus $\boxed{q(x) = 6x^4 + 7x^3 + 2x^2 + 10x + 2}$ and $\boxed{r(x) = 4}$. The final remainder has degree below the divisor's degree; reversing the subtractions verifies $f = gq + r$.

Exercise 23.4: Polynomial division 4

Work in $\mathbb{Z}_{11}[x]$, reducing coefficients after each subtraction. The inverse of the divisor's leading coefficient is 9. Repeatedly subtract the multiple that cancels the current highest power. The complete long-division steps are

term added to quotient	new remainder
$9x^2$	$3x^3 + x^2$
$5x$	$6x^2 + x$
10	2

Thus $\boxed{q(x) = 9x^2 + 5x + 10}$ and $\boxed{r(x) = 2}$. The final remainder has degree below the divisor's degree; reversing the subtractions verifies $f = gq + r$.

Exercise 23.5: All multiplicative generators modulo 5

The group has order 4. The element 2 has this full order: it satisfies $2^4 = 1$, and the checks $2^2 \equiv 4 \pmod 5$ exclude every proper divisor of the order (check exponents $(4)/q$ for each prime q dividing 4). In a cyclic group of order 4, the generators are exactly g^k with $\gcd(k, 4) = 1$. Here those exponents are $1, 3$, yielding the complete generator set

$$\boxed{\{2, 3\}}.$$

Exercise 23.6: All multiplicative generators modulo 7

The group has order 6. The element 3 has this full order: it satisfies $3^6 = 1$, and the checks $3^3 \equiv 6, 3^2 \equiv 2 \pmod 7$ exclude every proper divisor of the order (check exponents $(6)/q$ for each prime q dividing 6). In a cyclic group of order 6, the generators are exactly g^k with $\gcd(k, 6) = 1$. Here those exponents are $1, 5$, yielding the complete generator set

$$\boxed{\{3, 5\}}.$$

Exercise 23.7: All multiplicative generators modulo 17

The group has order 16. The element 3 has this full order: it satisfies $3^{16} = 1$, and the checks $3^8 \equiv 16 \pmod{17}$ exclude every proper divisor of the order (check exponents $(16)/q$ for each prime q dividing 16). In a cyclic group of order 16, the generators are exactly g^k with $\gcd(k, 16) = 1$. Here those exponents are $1, 3, 5, 7, 9, 11, 13, 15$, yielding the complete generator set

$$\boxed{\{3, 5, 6, 7, 10, 11, 12, 14\}}.$$

Exercise 23.8: All multiplicative generators modulo 23

The group has order 22. The element 5 has this full order: it satisfies $5^{22} = 1$, and the checks $5^{11} \equiv 22, 5^2 \equiv 2 \pmod{23}$ exclude every proper divisor of the order (check exponents $(22)/q$ for each prime q dividing 22). In a cyclic group of order 22, the generators are exactly g^k with $\gcd(k, 22) = 1$. Here those exponents are $1, 3, 5, 7, 9, 13, 15, 17, 19, 21$, yielding the complete generator set

$$\boxed{\{5, 7, 10, 11, 14, 15, 17, 19, 20, 21\}}.$$

Exercise 23.9: Four linear factors modulo five

Since $4 = -1$ in $\mathbb{Z}_5$,

$$x^4 + 4 = x^4 - 1 = (x^2 - 1)(x^2 + 1).$$

Now $x^2 - 1 = (x - 1)(x + 1)$ and $x^2 + 1 = x^2 - 4 = (x - 2)(x + 2)$. Therefore the requested linear factorization is

$$\boxed{(x - 1)(x + 1)(x - 2)(x + 2)}.$$

Exercise 23.10: Three linear factors modulo seven

Substitution shows that $x = -1$ is a zero. Division gives

$$x^3 + 2x^2 + 2x + 1 = (x + 1)(x^2 + x + 1).$$

In $\mathbb{Z}_7$, $(x - 2)(x + 3) = x^2 + x - 6 = x^2 + x + 1$. Thus

$$\boxed{x^3 + 2x^2 + 2x + 1 = (x + 1)(x - 2)(x + 3)}.$$

Expansion checks all coefficients modulo 7.

Exercise 23.11: A nonmonic cubic modulo eleven

The residues $3, 4, -3$ are zeros, suggesting the three factors. Their product expands as

$$2(x - 4)(x - 3)(x + 3) = 2(x - 4)(x^2 - 9) = 2x^3 - 8x^2 - 18x + 72.$$

Reducing modulo 11 gives $2x^3+3x^2-7x-5$, as required. Hence this is the complete linear factorization, including its leading coefficient 2.

Exercise 23.12: A reducible cubic modulo five

The value at $x = 2$ is $8 + 4 + 3 = 15 = 0$ modulo 5, so the cubic is not irreducible. Division by $x - 2$ gives $x^2 + 2x + 1 = (x + 1)^2$. Therefore

$$\boxed{x^3 + 2x + 3 = (x - 2)(x + 1)^2} \quad \text{in } \mathbb{Z}_5[x].$$

All three linear factors are irreducible, with a repeated factor explicitly recorded.

Exercise 23.13: An irreducible cubic modulo five

A cubic over a field is reducible exactly when it has a zero in the field. The values of $2x^3 + x^2 + 2x + 2$ at $0, 1, 2, 3, 4$ are $2, 2, 1, 1, 4$ in $\mathbb{Z}_5$, none zero. It is therefore irreducible. If a monic irreducible factor is desired, write it as

$$\boxed{2(x^3 + 3x^2 + x + 1)}.$$

The scalar 2 is a unit, and the cubic factor is irreducible; there is no factorization into positive lower degrees.

Exercise 23.14: A quadratic over three fields

The discriminant of $x^2 + 8x - 2$ is $8^2 - 4(1)(-2) = 72$. Its roots are $-4 \pm 3\sqrt{2}$, neither rational. A quadratic over a field is reducible if and only if it has a root there, so it is irreducible over $\mathbb{Q}$. Both roots are real, however, giving

$$x^2 + 8x - 2 = (x + 4 - 3\sqrt{2})(x + 4 + 3\sqrt{2}).$$

Thus it is reducible over $\mathbb{R}$ and consequently over $\mathbb{C}$.

Exercise 23.15: A quadratic with negative discriminant

For $x^2 + 6x + 12$, the discriminant is $36 - 48 = -12$. There is no real root, hence no rational root. It is irreducible over both $\mathbb{Q}$ and $\mathbb{R}$. Over $\mathbb{C}$, its roots are $-3 \pm i\sqrt{3}$, so

$$x^2 + 6x + 12 = (x + 3 - i\sqrt{3})(x + 3 + i\sqrt{3})$$

is a factorization and it is reducible there.

Exercise 23.16: A cubic over the rationals

A reducible cubic over $\mathbb{Q}$ would have a rational root. The rational-root theorem for the monic polynomial $x^3 + 3x^2 - 8$ limits candidates to $\pm 1, \pm 2, \pm 4, \pm 8$. Their values, in that order, are

$$-4, -6, 12, -4, 104, -24, 696, -328,$$

all nonzero. Thus there is no rational root and the cubic is irreducible over $\mathbb{Q}$.

Exercise 23.17: A quartic with no quadratic factor

The only possible rational roots of the monic integer polynomial are ± 1, and both give -20, so no linear factor exists. By Gauss's lemma, any remaining rational factorization could be taken as

$$(x^2 + ax + b)(x^2 + cx + d), \qquad a, b, c, d \in \mathbb{Z}.$$

Comparing the x^3 and constant coefficients gives $c = -a$ and $bd = 1$, so $(b, d) = (1, 1)$ or $(-1, -1)$. The x^2 coefficient is $b + d - a^2 = -22$. In the first case this forces $a^2 = 24$; in the second it forces $a^2 = 20$. Neither is an integer square. Thus there is no quadratic factorization either, and $x^4 - 22x^2 + 1$ is irreducible over $\mathbb{Q}$.

Exercise 23.18: An Eisenstein quadratic

For $x^2 - 12$, use the prime $p = 3$. It does not divide the leading coefficient 1, divides all other coefficients (including the zero linear coefficient), and $3^2 = 9$ does not divide -12. Eisenstein's criterion therefore applies and proves irreducibility over $\mathbb{Q}$.

Exercise 23.19: An Eisenstein cubic

Use $p = 3$: it does not divide the leading coefficient 8, it divides the other coefficients $6, -9, 24$, and 9 does not divide the constant coefficient 24. Thus all Eisenstein hypotheses hold and the cubic is irreducible over $\mathbb{Q}$.

Exercise 23.20: An Eisenstein polynomial of degree ten

Use $p = 3$ for $4x^{10} - 9x^3 + 24x - 18$. Although $3 \nmid 4$ and 3 divides all lower coefficients, $3^2 = 9$ divides the constant coefficient -18. This prime fails the square condition. No other prime divides all the lower nonzero coefficients, since their gcd is 3. Thus the polynomial satisfies no Eisenstein criterion in its displayed form. Failure of the criterion alone does not decide reducibility.

Exercise 23.21: Another Eisenstein polynomial of degree ten

For $2x^{10} - 25x^3 + 10x^2 - 30$, the prime 5 divides every lower coefficient but not the leading coefficient 2. Its square 25 does not divide -30. Thus Eisenstein applies with $p = 5$, proving irreducibility over $\mathbb{Q}$.

Exercise 23.22: All rational zeros of a quartic

The rational-root theorem permits only fractions with numerator dividing 10 and denominator dividing 6. Testing candidates finds $x = -5/2$ and $x = 2/3$. Exact division, or multiplication, gives

$$6x^4 + 17x^3 + 7x^2 + x - 10 = (2x + 5)(3x - 2)(x^2 + x + 1).$$

The final quadratic has discriminant $1 - 4 = -3$, so it has no rational zero. Consequently the complete rational zero set is $\boxed{\{-5/2,\ 2/3\}}$. The displayed factorization checks the candidate search and excludes further rational roots.

Exercise 23.23: Correcting irreducibility

A nonconstant polynomial $f \in F[x]$ is irreducible over F if it cannot be written as $f = gh$ with *both* $g, h \in F[x]$ of positive degree. The printed definition forbids even $f = 1 \cdot f$, which every polynomial has. Trivial factorizations by nonzero constant units must be allowed.

Exercise 23.24: An equivalent irreducibility definition

The stated definition is correct. A nonconstant f is nonzero. In any factorization $f = gh$, both factors are nonzero, so a factor belonging to F is a nonzero constant and hence a unit in $F[x]$. Requiring such a factor in every factorization is equivalent to forbidding a product of two positive-degree polynomials.

Exercise 23.25: True or false: irreducibility and root bounds

(a) True: a degree-one polynomial cannot be a product of two positive-degree polynomials.

(b) True for the same reason; the nonzero scalar factor 3 is a unit in $\mathbb{Q}[x]$.

(c) True: $\sqrt{3}$ is irrational, so this quadratic has no rational root.

(d) False: in $\mathbb{Z}_7$, $2^2 + 3 = 7 = 0$, so $x^2 + 3$ is reducible.

(e) True: degree additivity forces any polynomial unit over a field to be a nonzero constant.

(f) True: the source repeats the same assertion as in (e), so the same reasoning applies.

(g) True: a polynomial of degree n is nonzero and has at most n distinct zeros over a field.

(h) True: regard it as the same nonzero degree-n polynomial over the extension field E and apply the root bound there.

(i) True: $ax + b$, $a \neq 0$, has zero $-b/a \in F$.

(j) False: over an infinite field such as $\mathbb{Q}$, the zero polynomial vanishes at infinitely many inputs. The assertion becomes true if one requires the polynomial to be nonzero.

Exercise 23.26: Primes for which a specified linear factor divides

The factor theorem says that $x + 2 = x - (-2)$ divides the polynomial modulo p exactly when its value at -2 is zero modulo p. This value is

$$16 - 8 + 4 + 2 + 1 = 15.$$

Thus p must divide 15. The prime possibilities are exactly $\boxed{3, 5}$, and both do satisfy the condition.

Exercise 23.27: Irreducible quadratics over two elements

Every monic quadratic has form $x^2 + ax + b$. It has no root at 0 only if $b = 1$; then its value at 1 is a, which is nonzero only if $a = 1$. Thus the only irreducible quadratic is $\boxed{x^2 + x + 1}$. There is no further nonzero scalar multiple to list because 1 is the only nonzero scalar in $\mathbb{Z}_2$.

Exercise 23.28: Irreducible cubics over two elements

A cubic is irreducible exactly when it has no field root. Write it $x^3 + ax^2 + bx + c$. Avoiding root 0 requires $c = 1$; avoiding root 1 requires $a + b \neq 0$, so $(a, b) = (1, 0)$ or $(0, 1)$. The complete list is

$$\boxed{x^3 + x^2 + 1, \qquad x^3 + x + 1}.$$

Exercise 23.29: Irreducible quadratics over three elements

Test the monic quadratics $x^2 + ax + b$ at $0, 1, 2$. The monic ones with no zero are exactly

$$x^2 + 1, \quad x^2 + x + 2, \quad x^2 + 2x + 2.$$

Their respective value triples are $(1, 2, 2)$, $(2, 1, 2)$, and $(2, 2, 1)$. All quadratic leading coefficients are 1 or 2, and multiplying by a nonzero constant preserves irreducibility. Thus the full answer is these three polynomials and twice each of them, six in total.

Exercise 23.30: Irreducible cubics over three elements

A cubic is irreducible exactly when none of $0, 1, 2$ is a zero. The monic list, together with values at these inputs, is

$h(x)$	$h(0)$	$h(1)$	$h(2)$
$x^3 + 2x^2 + 1$	1	1	2
$x^3 + 2x^2 + x + 1$	1	2	1
$x^3 + 2x + 1$	1	1	1
$x^3 + x^2 + 2x + 1$	1	2	2
$x^3 + x^2 + 2$	2	1	2
$x^3 + x^2 + x + 2$	2	2	1
$x^3 + 2x + 2$	2	2	2
$x^3 + 2x^2 + 2x + 2$	2	1	1

.

To see completeness, write a monic cubic as $x^3 + ax^2 + bx + c$. Necessarily $c = 1$ or 2; testing the nine pairs (a, b) for each c and discarding zeros at 1 or 2 gives precisely these eight rows. The full list of irreducible cubics consists of these eight polynomials and their multiples by 2, sixteen in all.

Exercise 23.31: Counting irreducible quadratics over a prime field

There are p^2 monic quadratics. A reducible monic quadratic is uniquely $(x-a)(x-b)$ for an unordered pair of field elements, allowing $a = b$. There are $\binom{p}{2} + p = p(p+1)/2$ such pairs. Therefore the number of monic irreducible quadratics is

$$p^2 - \frac{p(p+1)}{2} = \frac{p(p-1)}{2}.$$

Each has $p - 1$ distinct nonzero scalar multiples, and every nonmonic quadratic is a unique scalar times a monic one. Thus the requested count of *all* irreducible quadratics is $\boxed{p(p-1)^2/2}$.

Exercise 23.32: Synopsis of the bound on roots

Each distinct zero supplies a distinct linear factor by the factor theorem, and repeatedly removing these factors lowers the degree by one each time; a nonzero degree-n polynomial cannot supply more than n such factors.

Exercise 23.33: Synopsis of cyclicity of finite field subgroups

By the finite abelian group theorem, write the finite multiplicative subgroup with invariant factors $d_1 \mid \cdots \mid d_r$. Every element then satisfies $x^{d_r} = 1$. The polynomial root bound gives $|G| \leq d_r$, while the product of invariant factors is $|G| \geq d_r$. Equality forces all earlier factors to be trivial, so the group is cyclic.

Exercise 23.34: A reducible prime-power polynomial

In $\mathbb{Z}_p$, Fermat gives $a^p = a$, and all intermediate binomial coefficients of exponent p vanish. Hence

$$(x + a)^p = x^p + a^p = x^p + a.$$

Since $p \geq 2$, this is a product of p linear factors, so it is not irreducible. This includes $a = 0$, when the polynomial is x^p.

Exercise 23.35: The reciprocal polynomial

Let $g(x) = a_n + a_{n-1}x + \cdots + a_0x^n$. Since $a \neq 0$, substitute a^{-1} and factor out a^{-n}:

$$g(a^{-1}) = a^{-n}(a_0 + a_1a + \cdots + a_na^n) = a^{-n}f(a) = 0.$$

Thus $1/a$ is a zero of the reciprocal polynomial, with the nonzero assumption justifying all inverse powers.

Exercise 23.36: The remainder theorem

Division by $x - a$ gives $f(x) = (x - a)q(x) + r(x)$, where $r = 0$ or $\deg r < 1$. Thus r is a constant. Evaluating at a yields $f(a) = 0 \cdot q(a) + r = r$. Hence the remainder is exactly the constant polynomial $\boxed{f(a)}$.

Exercise 23.37: Reduction of integer coefficients

(a) Reduce each coefficient modulo m. Sums are preserved coefficientwise; products are preserved because each coefficient of a product is a finite sum of coefficient products, and integer reduction preserves both operations. Every polynomial over $\mathbb{Z}_m$ can be lifted by choosing integer representatives for its finitely many coefficients, so the map is onto.

(b) Suppose instead that f is reducible over $\mathbb{Q}$. By Gauss's lemma it has a factorization $f = gh$ in $\mathbb{Z}[x]$ with $0 < \deg g, \deg h < n$. The leading coefficient of f remains nonzero modulo m by the degree hypothesis. Because it is the product of the leading coefficients of g, h, neither of those can become zero modulo m. Their reduced degrees are therefore unchanged. Reducing $f = gh$ then gives the forbidden factorization into two degrees below n, a contradiction. Thus f is irreducible over $\mathbb{Q}$.

(c) Modulo 5, the polynomial becomes $x^3 + 2x + 1$. Its values at $0, 1, 2, 3, 4$ are $1, 4, 3, 4, 3$, none zero. As a cubic over the field $\mathbb{Z}_5$, it is irreducible, and its degree remains 3. Part (b) proves that $x^3 + 17x + 36$ is irreducible over $\mathbb{Q}$.